

Claim that did not survive		Why it looked true	Diagnostic that exposed it
R1	D2 cation swaps cannot be detected (3–14% over four searches)	Four groups of structural quantities all failed and two physical explanations seemed plausible	Madelung energy of every damaged cell was exactly zero because the damage construction was flawed
R2	Law thresholds were fitted only on the discovery split	The split file existed and was cited in the written protocol	After the analysis set grew, 3,215 rows without split labels entered one fit
R3	The satisfaction–damage-detection trade-off is a physical limit	The pattern recurred across six targets, four input groups and three analysis levels	The search used its full allowed loss of experimental-crystal satisfaction on the first law
R4	No unrestricted model can exceed 87% damage detection at 99% experimental-structure satisfaction	A boosted tree gave this result on separate parent structures	When each damage type was omitted during fitting, damage detection fell by 0.45–0.87
R5	Symmetry is the best single law (99.17% satisfaction, 39.5% damage detection)	Both numbers exceeded those of the core law	Damage detection was 0.98 for random displacement and 0.00 for expansion: a pattern of the damage procedure rather than chemistry
R6	Charge-topology quantities add nothing because removing them changes no result	The repeated analysis matched to the last digit	The removal switch deleted only unconditional laws and left the conditional versions in place
R7	Materials Project labels are noisier because the database is heterogeneous	Accuracy rose steadily with energy difference and adding label noise reproduced the trend	Three databases computed with consistent settings all scored below Materials Project in matched energy ranges
R8	Structural criteria predict energy ordering (0.6603 = 0.6603, exact)	The two values agreed to four decimal places	On the subset ranked correctly by energy the two quantities are equal by definition
R9	Density predicts synthesis history better than formation energy	0.7103 versus 0.6476, unchanged by four checks	SiO2 supplied 43.6% of all pairs. Giving each composition equal weight reversed the result to 0.6347 versus 0.7451
R10	Pauling's rules perform worse than chance (0.14–0.36)	The values were far below chance and seemed to have a symmetry-based explanation	Ties made up 77.7% of pairs and were counted as errors. Experimentally recorded entries were actually more symmetric (space group 87 versus 62)
R11	An upper contact bound adds no value because the law must be one-sided	At 99% satisfaction the best range is [0.73, 3.04], so the upper bound detects no additional damage and damage detection (0.2823) is below the one-sided law (0.2881)	At equal satisfaction below 95%, the best upper bound changes sharply from about 3.0 to 1.08 and improves damage detection by up to 0.16